

Week 36, Tuesday Session

Flipped classroom: your questions, then bias-variance and cross-validation in practice

Bring your laptop – we work in `week36tuesday.ipynb`

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education,
University of Oslo, Norway

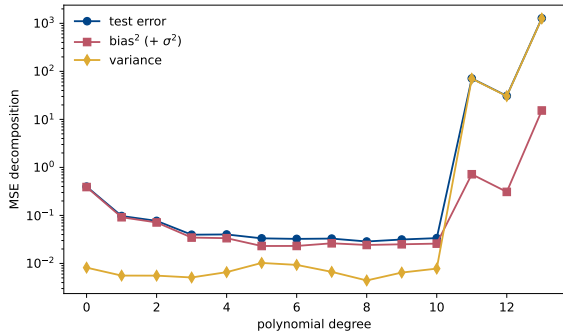
How today works

The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete cases. You have heard the lecture and skimmed Sections 2.3–2.12.

1. **First ~20 min:** open questions from Monday, then the warm-ups on the next slides, discussed with your neighbours and in plenum.
2. **Rest of the double session:** *two* cases only this week – the bias-variance decomposition, and cross-validation – worked in pairs or small groups from `week36tuesday.ipynb`. Deep beats broad: the goal is that you leave owning Eq. (2.47) and a working k -fold loop, both of which Project 1 asks for.
3. **Last 5 min:** wrap-up, exercise deadline, and what next Monday builds on.

Picking up from Monday: the measured decomposition



Monday's lecture closed the loop on last week's U-curve: the test error is, by Eq. (2.47), the sum

$$\text{Bias}^2 + \text{Var} + \sigma^2,$$

and the bootstrap let us *measure* each term. Bias falls and flattens, variance explodes, the sum has a broad minimum (degrees 5–9 here). Today you reproduce this figure

yourselves, stress-test it – more data, more noise, a Ridge penalty – and then let cross-validation find the minimum without ever touching the test set.

Warm-up 1: the three terms, in your own words

Question

Without looking at your notes: which term of Eq. (2.47) does each sentence describe? (a) “No amount of data fixes a model that is too simple.” (b) “Redraw the training set and the prediction jumps.” (c) “Even the perfect model gets this wrong.”

Warm-up 1: the three terms, in your own words

Question

Without looking at your notes: which term of Eq. (2.47) does each sentence describe? (a) “No amount of data fixes a model that is too simple.” (b) “Redraw the training set and the prediction jumps.” (c) “Even the perfect model gets this wrong.”

Answer

(a) the squared bias – the error of the *average* prediction, built into the model’s assumptions; (b) the variance – sensitivity to the particular training sample; (c) the irreducible σ^2 – the noise no model predicts. Bonus check: which one does Ridge attack? The variance, at the price of some bias.

Warm-up 2: read the code, not the formula

Question

In Monday's bootstrap code the bias is computed as `np.mean((y_test - np.mean(y_pred, axis=1, keepdims=True))*2)`. Two questions: why `axis=1`, and why does this "bias" contain σ^2 ?

Warm-up 2: read the code, not the formula

Question

In Monday's bootstrap code the bias is computed as `np.mean((y_test - np.mean(y_pred, axis=1, keepdims=True))*2)`. Two questions: why `axis=1`, and why does this “bias” contain σ^2 ?

Answer

Columns of `y_pred` are bootstrap training sets, so averaging over `axis=1` is the expectation over training sets, $\mathbb{E}[\tilde{\mathbf{y}}]$ – averaging over `axis=0` would average over test points instead, which is meaningless here. And the code compares with $y_{\text{test}} = f + \varepsilon$, not the unknown f : the noise cannot be subtracted out, so the measured bias absorbs σ^2 and the printed identity is really an inequality.

Warm-up 3: predict before you compute

Question

Case 1 below reruns the bias-variance sweep with (a) $n = 400$ instead of 40 data points, and (b) noise $\sigma = 0.3$ instead of 0.1. Predict: what happens to each of the three curves in each case?

Warm-up 3: predict before you compute

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Answer

(a) More data tames the variance (roughly like $1/n$ for fixed complexity): the explosion moves to higher degree and the minimum shifts right – with more data you can afford a richer model. The bias curve is unchanged: it is a property of the model family, not of n . (b) More noise lifts the floor (σ^2 is 9 times larger) and scales up the variance term; the bias² term is unchanged, but the minimum moves *left*: noisier data favour simpler models. Now go check.

Case 1: the bias-variance decomposition

Goal: reproduce Monday's figure, then break it and understand the pieces (notebook, Case 1; seed 2026 gives exactly the lecture numbers).

1. Run the sweep ($n = 40, 100$ bootstraps, degrees 0–13) and reproduce `week36_bias_variance.pdf`.
2. **Vary the data size:** $n = 100$ and $n = 400$. Where does the variance explosion move? Does the bias curve care?
3. **Vary the noise:** $\sigma = 0.3$. Confirm your warm-up 3 prediction for floor and minimum.
4. **Swap the model:** replace OLS by Ridge with $\lambda \in \{10^{-4}, 10^{-1}, 10\}$ at fixed high degree 12. Watch the variance fall and the bias rise as λ grows – the tradeoff turned into a dial.

Checkpoints (plenary after ~ 25 min)

Degree 0 has $\text{bias}^2 \approx 0.39$ ($\approx$ the variance of f itself – why?); the minimum is broad, degrees 5–9; at degree 13 the variance is $\sim 10^3$. For step 4: which λ gives the smallest *total* error at degree 12, and is it better than plain OLS at degree 6?

Case 2: cross-validation: choosing λ without touching the test set

Goal: a k -fold loop using `scikit-learn` or your own code, then used for real model selection (notebook, Case 2).

1. Implement 5-fold CV for Ridge on $y = 3x^2 + \varepsilon$ (degree-6 polynomial, λ on `np.logspace(-3, 5, 100)`), scaler fitted *inside* each fold. Reproduce `week36_cv_ridge.pdf`.
2. Read off λ^* and compare the minimal CV-MSE with the known $\sigma^2 = 1$. Vary $k \in \{5, 10, 20\}$: how stable are λ^* and the error bar (the spread over folds)?
3. **Close the week's loop:** run CV over the polynomial *degree* for the week-35 data set (OLS, degrees 0–13) and compare the CV curve with Case 1's test-error curve – same minimum, no test set consumed.

Checkpoints

With seed 2026: $\lambda^* \approx 1.2$, CV-MSE ≈ 1.5 (just above the noise floor). For step 3: CV picks a degree in the same broad 5–9 window as the bootstrap decomposition, that is two independent methods, one answer.

Wrap-up, deadlines, and Monday

- ▶ **What you should own now:** Eq. (2.47) and what each term means operationally; a bootstrap loop; a leak-free k -fold loop with the preprocessing inside the folds.
- ▶ **Weekly exercises** (`exercisesweek36.ipynb`): the analytical steps of the bias-variance derivation, plus bootstrap and cross-validation on your own code. Deadline **Friday September 4 at midnight**.
- ▶ **Project 1** asks for exactly today's machinery (bias-variance analysis and cross-validation on regression models) – the code you wrote today is directly reusable.
- ▶ **Next Monday and week 38:** the main focus shifts to *optimization*: gradient descent methods, from plain and steepest descent via momentum to stochastic gradient descent. The latter is the workhorse behind everything from Lasso to deep networks. We will also briefly mention the Bayesian reading of OLS, Ridge and Lasso (a Gaussian prior gives Ridge, a Laplace prior the Lasso).

Before you leave

One line in an email to us: the muddiest point of the week, what should Monday's first ten minutes next week revisit?